DYNAMICAL SYSTEMS
PROBLEM SET 4

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We begin by noting that all of the files referenced in this document are available at:
https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet4

8.1.6 Consider the system $\dot{x} = y - 2x$, $\dot{y} = \mu + x^2 - y$.

a) Sketch the nullclines.
b) Find and classify the bifurcations that occur as μ varies.
c) Sketch the phase portrait as a function of μ.

Solution. **(a) Nullclines.** Setting each component of the vector field equal to zero gives

$$\dot{x} = 0: \quad y = 2x, \qquad \dot{y} = 0: \quad y = x^2 + \mu.$$

Thus the nullclines are a fixed straight line and an upward-opening parabola that moves vertically with μ. The direction of the flow is determined by

	condition	direction
$\dot{x} > 0$	$y > 2x$	right
$\dot{x} < 0$	$y < 2x$	left
$\dot{y} > 0$	$y < x^2 + \mu$	up
$\dot{y} < 0$	$y > x^2 + \mu$	down

The nullclines intersect when

$$x^2 - 2x + \mu = (x-1)^2 + \mu - 1 = 0.$$

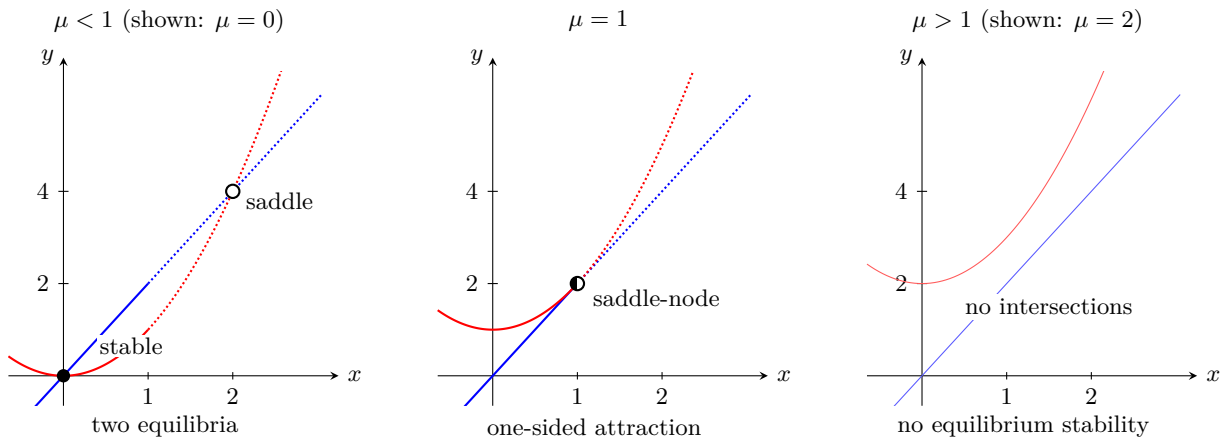
For $\mu < 1$, the two equilibria are

$$P_- = \big(1 - \sqrt{1-\mu},\, 2 - 2\sqrt{1-\mu}\big), \qquad P_+ = \big(1 + \sqrt{1-\mu},\, 2 + 2\sqrt{1-\mu}\big).$$

At $\mu = 1$ they coincide at $(1, 2)$, where the nullclines are tangent; for $\mu > 1$ there are no equilibria.

Nullcline sketches and equilibrium stability. The following sketches use $\mu = 0, 1, 2$ as representatives of the three cases. Blue denotes $y = 2x$ and red denotes $y = x^2 + \mu$. For $\mu \leq 1$, solid portions on the $x < 1$ side indicate the attracting equilibrium branch, and dotted portions on the $x > 1$ side indicate the saddle branch or the repelling side of the saddle-node. These line styles are visual guides to the classification at the intersections: points elsewhere on a nullcline are not equilibria and do not themselves have a stability classification. In particular, the dotted branch at the saddle does not mean that both eigen-directions are unstable.

Date: November 4, 2026.



At $\mu = 1$, "one-sided attraction" refers to the center-manifold dynamics: the equilibrium attracts from the left and repels to the right, while the transverse direction is stable. It is therefore not a stable equilibrium in the Lyapunov sense. For $\mu > 1$, the curves are shown thin because there is no equilibrium to classify; calling either nullcline stable or unstable would be misleading.

(b) Jacobian and stability. The Jacobian is

$$J(x, y) = \begin{pmatrix} -2 & 1 \\ 2x & -1 \end{pmatrix}.$$

At the two equilibria, substitute $x = 1 \pm \sqrt{1-\mu}$ directly into the Jacobian:

$$J(P_\pm) = \begin{pmatrix} -2 & 1 \\ 2 \pm 2\sqrt{1-\mu} & -1 \end{pmatrix}.$$

Their traces, determinants, and characteristic polynomials are

$$\tau_\pm = -3, \qquad \Delta_\pm = -\big(\pm 2\sqrt{1-\mu}\big),$$

$$\det(\lambda I - J(P_\pm)) = \lambda^2 + 3\lambda - \big(\pm 2\sqrt{1-\mu}\big).$$

Here the upper sign corresponds to P_+ and the lower sign to P_-. We examine the two branches separately below. At the lower equilibrium P_-,

$$J_- = \begin{pmatrix} -2 & 1 \\ 2 - 2\sqrt{1-\mu} & -1 \end{pmatrix}, \qquad \Delta_- = 2\sqrt{1-\mu} > 0.$$

$$\lambda_{1,2}^{(-)} = \frac{-3 \pm \sqrt{9 - 8\sqrt{1-\mu}}}{2}.$$

Since the trace is negative and the determinant is positive, both eigenvalues have negative real parts. Hence P_- is asymptotically stable for every $\mu < 1$. More precisely,

parameter range	eigenvalues at P_-	classification
$\mu < -17/64$	complex, real part $-3/2$	stable spiral
$\mu = -17/64$	$\lambda_1 = \lambda_2 = -3/2$	improper stable node
$-17/64 < \mu < 1$	distinct, real, negative	stable node

Indeed, $9 - 8\sqrt{1-\mu} = 0$ gives $\sqrt{1-\mu} = 9/8$ and $\mu = 1 - (9/8)^2 = -17/64$. At this value $J_- \neq -\frac{3}{2}I$, so the repeated eigenvalue has only one independent eigenvector. This spiral-to-node transition does not change stability and is not a bifurcation of the local topological phase portrait.

At the upper equilibrium P_+,

$$J_+ = \begin{pmatrix} -2 & 1 \\ 2 + 2\sqrt{1-\mu} & -1 \end{pmatrix}, \qquad \Delta_+ = -2\sqrt{1-\mu} < 0.$$

$$\lambda_{1,2}^{(+)} = \frac{-3 \pm \sqrt{9 + 8\sqrt{1-\mu}}}{2}.$$

One eigenvalue is positive and one is negative, so P_+ is always an unstable saddle.

At $\mu = 1$,

$$J(1,2) = \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix}, \qquad \lambda_1 = 0, \quad \lambda_2 = -3.$$

Linearization alone cannot classify the zero-eigenvalue direction. Set

$$u = x - 1, \qquad w = y - 2x, \qquad \delta = \mu - 1.$$

Then the system becomes exactly

$$\dot{u} = w, \qquad \dot{w} = \delta + u^2 - 3w.$$

At $\delta = 0$, a center manifold $w = h(u)$ satisfies

$$h'(u)h(u) = u^2 - 3h(u).$$

Writing $h(u) = au^2 + O(u^3)$ and comparing quadratic terms gives $a = 1/3$. Thus the reduced equation is

$$\dot{u} = \tfrac{1}{3}u^2 + O(u^3).$$

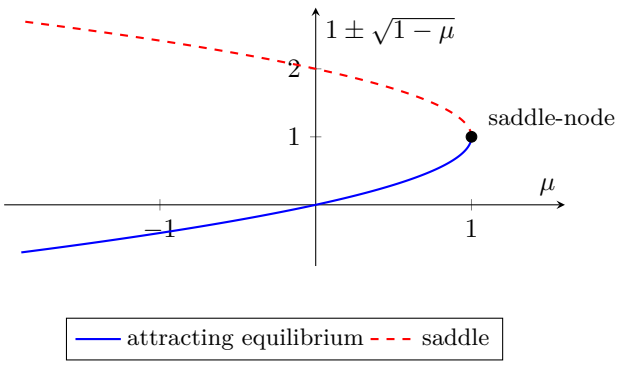
The equilibrium attracts along the center manifold from $u < 0$ and repels toward $u > 0$, while the transverse direction is attracting. In particular, it is not Lyapunov stable. Including the parameter gives the local saddle-node form

$$\dot{u} = \tfrac{1}{3}(\delta + u^2) + O\big(|u|^3 + |\delta u| + \delta^2\big).$$

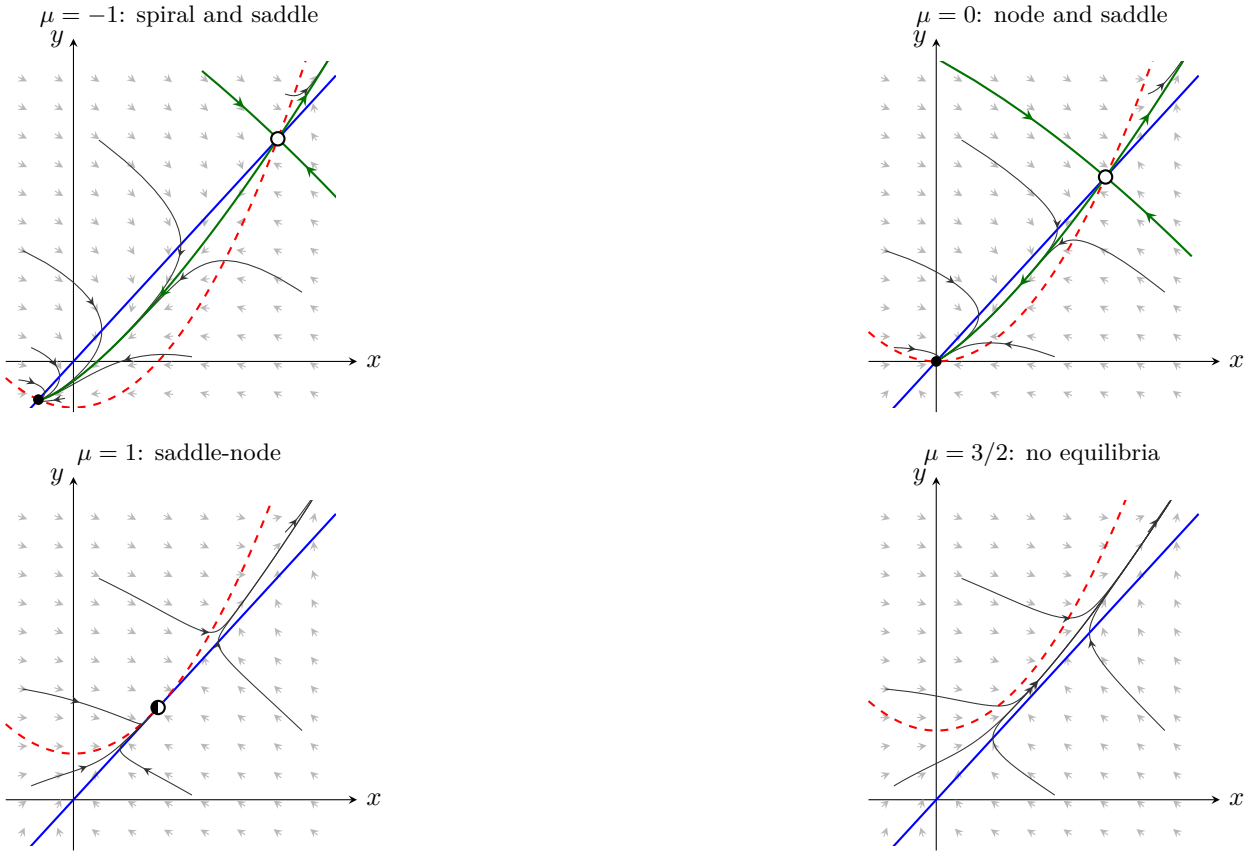
Therefore a stable equilibrium and a saddle collide and disappear at

$$\mu_c = 1 \quad \text{(saddle-node bifurcation)}.$$

No Hopf bifurcation is possible because the trace is always -3. Also, $\partial\dot{x}/\partial x + \partial\dot{y}/\partial y = -3$ everywhere, so Bendixson's criterion excludes periodic orbits in the plane.



(c) Phase portraits. The plots below show representative cases on either side of $\mu = 1$ and at the bifurcation. Blue solid curves are $\dot{x} = 0$, red dashed curves are $\dot{y} = 0$, and gray arrows give the flow direction. Black curves are numerically integrated trajectories; their arrows indicate increasing time. At the saddle, the green curves show its stable and unstable separatrices. A filled point marks the attractor, an open point the saddle, and a half-filled point the saddle-node.



For $\mu < 1$, nearby trajectories approach P_- (spiraling when $\mu < -17/64$), whereas the saddle's stable manifold separates trajectories heading toward the attractor from those escaping to the right. At $\mu = 1$, the two equilibria merge, leaving one-sided attraction and repulsion along the center direction. For $\mu > 1$ the parabola lies strictly above the line, and the flow passes through the region where the equilibria used to be. In the coordinates above,

$$\dot{w} = \mu - 1 + u^2 - 3w \geq \mu - 1 - 3w.$$

Thus, for any forward-complete trajectory, w is eventually positive and bounded away from zero, so $\dot{u} = w > 0$ eventually and u grows without bound. (Some trajectories may instead escape in finite time.) There is no remaining bounded attractor. Note: plots originally generated by file `ps4_8_1_6.m` □

8.1.15 (The power of true believers) Sometimes a small band of unwavering advocates can win an entire population over to their point of view, as in the case of the civil rights or women's suffrage movements in the United States. Consider the following stylized model of such situations, which was studied by Marvel et al. (2012) and inspired by the earlier work of Xie et al. (2011).

The population is divided into four non-overlapping groups. An initially small group of true believers holds opinion A (for example, that women deserve the right to vote) and they are committed to this belief. Nothing

that anyone says or does can change their minds. Another group of people currently agrees with them, but they are uncommitted to A. If someone argues for the opposing position, B, an uncommitted A-believer instantly becomes an AB, meaning someone who sees the merit in both positions. Likewise, B-believers instantly turn into members of the AB subpopulation when confronted with an argument for A. The people in the AB group, being fence-sitters, don't try to persuade anyone of anything. And they can be pushed to either side with the slightest nudge; when confronted with an argument for A—or for B—they join that camp. At each time step, we select two people at random and have one of them act as an advocate and the other as a listener. Assuming the members of the four groups mix with each other at random, the governing equations for the dynamics are

$$\dot{n}_A = (p + n_A)n_{AB} - n_A n_B,$$
$$\dot{n}_B = n_B n_{AB} - (p + n_A)n_{AB}.$$

where $n_{AB} = 1 - (p + n_A) - n_B$. Here the parameter p denotes the unchanging fraction of true believers in the population. The time-dependent variables n_A, n_B, and n_{AB} are the current fractions in the A, B, and AB subpopulations.

a) Interpret and justify the form of the various terms in the governing equations.
b) Assume that initially everyone believes in B, except for the true believers in A. Thus, $n_B(0) = 1 - p$, and $n_A(0) = n_{AB}(0) = 0$. Numerically integrate the system until it reaches equilibrium. Show that the final state changes discontinuously as a function of p. Specifically, show there is a critical fraction of true believers (call it p_c) such that for $p < p_c$, most people still accept B, whereas for $p > p_c$, everyone comes around to A.
c) Show analytically that $p_c = 1 - \sqrt{3}/2 \approx 0.134$. Thus, in this model, only about 13% of the population needs to be unwavering advocates to get everyone else to agree with them eventually.
d) What type of bifurcation occurs at p_c?

Solution. **Model used.** The governing equations are

$$\dot{n}_A = (p + n_A)n_{AB} - n_A n_B, \qquad \dot{n}_B = n_B n_{AB} - (p + n_A)n_B.$$

Thus the final factor in the second equation is n_B; the n_{AB} printed in the problem statement is a typographical error. All calculations below use this corrected equation.

(a) A listener in group AB becomes an A-believer when the speaker is either a committed or an uncommitted A-believer, giving the gain $(p+n_A)n_{AB}$. An A-listener becomes AB when the speaker is B, giving the loss $n_A n_B$. Similarly, a B-speaker converts an AB-listener to B at rate $n_B n_{AB}$, while an A-speaker converts a B-listener to AB at rate $(p + n_A)n_B$. AB-speakers do not persuade anyone. Conservation gives $n_{AB} = 1 - p - n_A - n_B$. The physical region is the invariant triangle

$$n_A \geq 0, \qquad n_B \geq 0, \qquad n_A + n_B \leq 1 - p.$$

On the three edges, respectively, $\dot{n}_A = p n_{AB} \geq 0$, $\dot{n}_B = 0$, and $\dot{n}_{AB} = n_B(p + 2n_A) \geq 0$.

(b) Numerical integration from $(n_A, n_B) = (0, 1 - p)$ shows an abrupt change in the final state: for example, $p = 0.10$ approaches a B-dominated equilibrium, whereas $p = 0.15$ approaches $(1 - p, 0)$. The time traces below also include $p = 0.13$ and $p = 0.14$, on opposite sides of the threshold. In the A-consensus state the committed fraction p and the uncommitted fraction $1 - p$ together comprise the entire population.

(c) From $\dot{n}_B = 0$, either $n_B = 0$, giving the consensus equilibrium $(1 - p, 0)$, or

$$n_B = 1 - 2p - 2n_A.$$

Substitution into $\dot{n}_A = 0$ yields

$$3n_A^2 + (4p-1)n_A + p^2 = 0.$$

The two additional equilibria therefore have

$$n_A = \frac{1 - 4p \pm \sqrt{4p^2 - 8p + 1}}{6}, \qquad n_B = 1 - 2p - 2n_A.$$

They collide when $4p^2 - 8p + 1 = 0$. Only the smaller root lies in $[0, 1]$, so

$$p_c = 1 - \frac{\sqrt{3}}{2} \approx 0.1339746.$$

To check stability directly, the Jacobian in coordinates (n_A, n_B) is

$$J = \begin{pmatrix} 1 - 2p - 2n_A - 2n_B & -p - 2n_A \\ -2n_B & 1 - 2p - 2n_A - 2n_B \end{pmatrix}.$$

At $(1 - p, 0)$,

$$J = \begin{pmatrix} -1 & p - 2 \\ 0 & -1 \end{pmatrix}, \qquad \lambda_1 = \lambda_2 = -1.$$

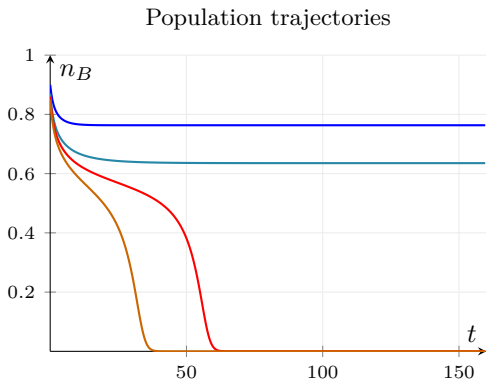
Thus consensus remains stable on both sides of the threshold. For a nonconsensus equilibrium, write $D = \sqrt{4p^2 - 8p + 1}$. The trace is $-2n_B < 0$, and the determinant is $n_B(1 - 4p - 6n_A)$. The root with the minus sign in n_A has determinant $n_B D > 0$ and is stable; the root with the plus sign has determinant $-n_B D < 0$ and is a saddle. The physical nonconsensus branches exist for $0 \le p \le p_c$. For $0 < p < p_c$, the initial condition in (b) approaches the B-dominated stable branch. For $p > p_c$, only A-consensus remains in the physical triangle.

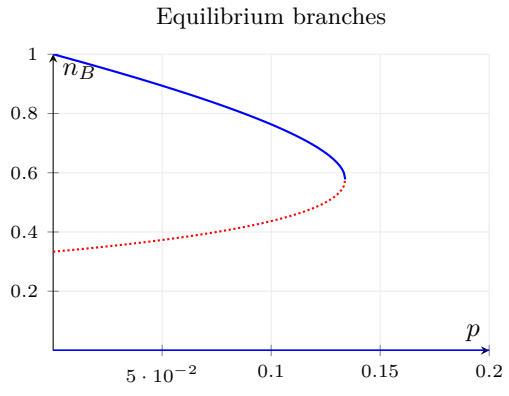
For $p > p_c$, convergence to the remaining equilibrium can also be justified beyond linearization. In coordinates $a = p + n_A$, $b = n_B$, choose the Dulac multiplier $1/(ab)$ on the interior of the physical triangle. Direct differentiation gives

$$\frac{\partial}{\partial a}\left(\frac{\dot{a}}{ab}\right) + \frac{\partial}{\partial b}\left(\frac{\dot{b}}{ab}\right) = -\frac{1}{b} - \frac{1}{a} - \frac{p}{a^2} < 0.$$

Thus there are no interior periodic orbits; together with invariance and the boundary flow, this rules out an alternative recurrent attractor when consensus is the only equilibrium.

(d) The B-dominated attractor and the intervening saddle disappear in a saddle-node bifurcation at p_c. The discontinuity in the long-time outcome is therefore not a loss of stability of A-consensus.





Time traces: blue $p = 0.10$, teal 0.13, red 0.14, orange 0.15. In the branch diagram solid curves are stable and dotted curves are saddles. The consensus branch $n_B = 0$ stays stable.

MATLAB: `ps4_8_1_15.m`.

□

8.2.2 By calculating the linearization at the origin, show that the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ has pure imaginary eigenvalues when $\mu = 0$.

Solution. The origin is an equilibrium for every μ. Differentiating both equations gives

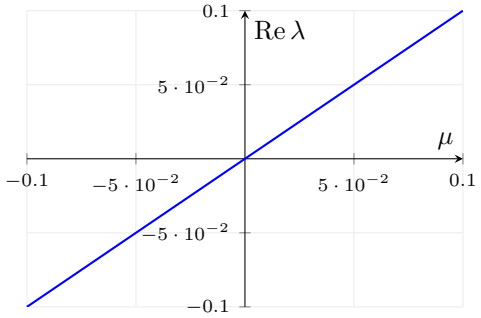
$$J(x,y) = \begin{pmatrix} \mu + y^2 & -1 + 2xy \\ 1 - 2x & \mu \end{pmatrix}, \qquad J(0,0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}.$$

The characteristic equation is

$$(\lambda - \mu)^2 + 1 = 0, \qquad \lambda_{1,2} = \mu \pm i.$$

At $\mu = 0$, the eigenvalues are the purely imaginary pair $\pm i$. For $\mu < 0$ the origin is a stable spiral; for $\mu > 0$ it is an unstable spiral. The eigenvalues cross the imaginary axis transversely because $d\,\mathrm{Re}\,\lambda/d\mu = 1$. At $\mu = 0$ the linearization is a center, but the nonlinear system is not a center; the nonlinear classification is established in Exercises 8.2.3 and 8.2.12.

Real parts of the eigenvalues



Both eigenvalues have real part μ; their imaginary parts are $+1$ and -1.

MATLAB: `ps4_8_2_2.m`.

□

8.2.3 (Computer work) By plotting phase portraits on the computer, show that the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ undergoes a Hopf bifurcation at $\mu = 0$. Is it subcritical, supercritical, or degenerate?

Solution. Compare phase portraits for negative, zero, and positive μ. The Jacobian calculation in 8.2.2 gives $\lambda = \mu \pm i$. When $\mu < 0$ is small, trajectories sufficiently near the origin spiral inward, but an unstable periodic orbit surrounds the origin: trajectories just outside it move away. This orbit shrinks to zero as $\mu \uparrow 0$. For $\mu > 0$ the origin is an unstable spiral and the nearby periodic orbit is absent. At $\mu = 0$ the origin is a weak unstable focus, with slow outward radial drift.

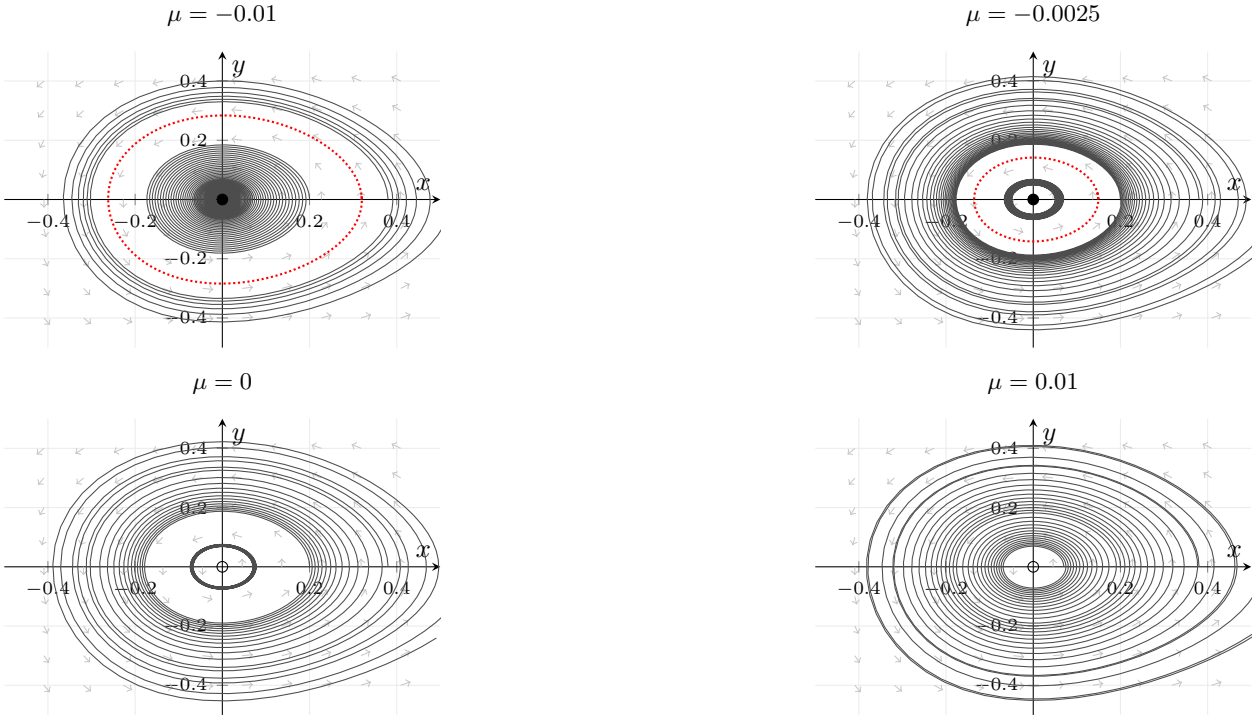
This is a *subcritical Hopf bifurcation*. The local radial normal form, confirmed analytically in 8.2.12, is

$$\dot{\rho} = \mu\rho + \frac{1}{8}\rho^3 + \text{higher-order terms}.$$

For $\mu < 0$, its nonzero equilibrium has $\rho \approx \sqrt{-8\mu}$ and radial derivative

$$\frac{d}{d\rho}\left(\mu\rho + \frac{1}{8}\rho^3\right)\Big|_{\rho^2 = -8\mu} = -2\mu > 0,$$

which explains the cycle's instability. Dotted closed curves in the negative-μ portrait below are numerically computed unstable cycles, not circular approximations. They can be traced by backward integration or by shooting on the section $y = 0$, $x > 0$.



Black curves are forward numerical trajectories; gray arrows show the vector field. Red dotted curves are unstable periodic orbits. The plotted window concerns the local Hopf dynamics only.

MATLAB: `ps4_8_2_3.m`.

□

8.2.4 (A heuristic analysis) The system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ can be analyzed in a rough, intuitive way as follows.

a) Rewrite the system in polar coordinates.

b) Show that if $r \ll 1$, then $\dot{\theta} \approx 1$ and $\dot{r} \approx \mu r + \frac{1}{8} r^3 + \cdots$, where the terms omitted are oscillatory and have essentially zero time-average around one cycle.

c) The formulas in part (b) suggest the presence of an unstable limit cycle of radius $r \approx \sqrt{-8\mu}$ for $\mu < 0$. Confirm that prediction numerically. (Since we assumed that $r \ll 1$, the prediction is expected to hold only if $|\mu| \ll 1$.)

The reasoning above is shaky. See Drazin (1992, pp. 188–190) for a proper analysis via the Poincaré-Lindstedt method.

Solution. **(a)** Put $x = r\cos\theta$, $y = r\sin\theta$. For $r > 0$,

$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r} = \mu r - r^2\cos^2\theta\sin\theta + r^3\cos^2\theta\sin^2\theta,$$

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1 - r\cos^3\theta - r^2\cos\theta\sin^3\theta.$$

(b) For $r \ll 1$, the angular speed is $1 + O(r)$. The relevant elementary averages are

$$\langle \cos^2\theta \sin\theta \rangle = 0, \qquad \langle \cos^2\theta \sin^2\theta \rangle = \frac{1}{8}.$$

Therefore the requested heuristic averaging gives

$$\dot{\theta} \approx 1, \qquad \dot{r} \approx \mu r + \frac{1}{8} r^3.$$

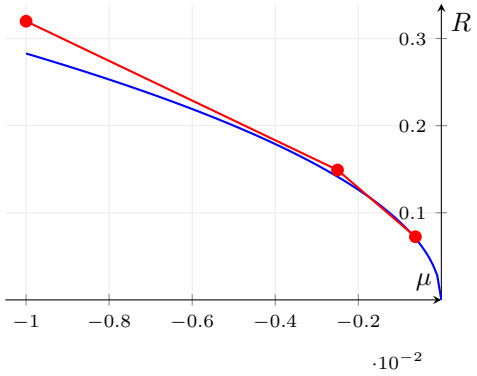
Here the equality is only approximate: the actual radius oscillates during each revolution, and simply discarding all zero-average terms is not a general proof of the normal form. The analytical criterion in 8.2.12 independently verifies the cubic coefficient's sign for this system.

(c) Solving the averaged radial equation gives $r \approx \sqrt{-8\mu}$ for $\mu < 0$. Its radial derivative there is $-2\mu > 0$, so the predicted orbit is unstable. For numerical confirmation, start at $(x, y) = (R, 0)$ and integrate to the next upward crossing of $y = 0$. If $P_\mu(R)$ denotes the new x-coordinate, solve $P_\mu(R) - R = 0$ by bisection. The following are section radii, not constant radii of circular orbits:

μ	$\sqrt{-8\mu}$	R from numerical shooting
-0.010000	0.282843	0.319862
-0.002500	0.141421	0.149129
-0.000625	0.070711	0.072497

The relative discrepancy decreases as $\mu \to 0^-$, as expected for a leading-order approximation. The phase portraits in 8.2.3 and the MATLAB file for this exercise show the corresponding periodic orbit and nearby trajectories.

Small-cycle section radius



Blue: leading prediction $\sqrt{-8\mu}$. Red points: section radii computed by shooting.

MATLAB: `ps4_8_2_4.m`.

□

8.2.10 (Bacterial respiration) Fairén and Velarde (1979) considered a model for respiration in a bacterial culture. The equations are

$$\dot{x} = B - x - \frac{xy}{1 + qx^2}, \qquad \dot{y} = A - \frac{xy}{1 + qx^2}$$

where x and y are the levels of nutrient and oxygen, respectively, and $A, B, q > 0$ are parameters. Investigate the dynamics of this model. As a start, find all the fixed points and classify them. Then consider the nullclines and try to construct a trapping region. Can you find conditions on A, B, q under which the system has a stable limit cycle? Use numerical integration, the Poincaré-Bendixson theorem, results about Hopf

bifurcations, or whatever else seems useful. (This question is deliberately open-ended and could serve as a class project; see how far you can go.)

Solution. Write $H(x) = x/(1+qx^2)$ and $d = B - A$. Subtracting the equations gives

$$\dot{x} - \dot{y} = d - x.$$

Hence every equilibrium must satisfy $x = d$. For $d \neq 0$, the unique real equilibrium is

$$(x, y) = \left(d, \frac{A(1+qd^2)}{d}\right).$$

If $d = 0$, there is no equilibrium, since $\dot{y} = A$ at $x = 0$. If $d < 0$, the equilibrium lies outside the physical quadrant and is a saddle. Thus a biologically relevant equilibrium exists exactly when $B > A$.

Jacobian and stability. At the equilibrium, put

$$c = \frac{d}{1+qd^2}, \qquad k = \frac{A(1-qd^2)}{d(1+qd^2)}.$$

Since $H'(x) = (1-qx^2)/(1+qx^2)^2$, the Jacobian is

$$J = \begin{pmatrix} -1-k & -c \\ -k & -c \end{pmatrix}, \qquad \tau = -1-k-c, \qquad \Delta = c,$$

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4c}}{2}.$$

For $d > 0$, the equilibrium is asymptotically stable when $\tau < 0$ and unstable when $\tau > 0$; it is a node if $\tau^2 > 4c$ and a spiral if $\tau^2 < 4c$. A repeated eigenvalue occurs when $\tau^2 = 4c$. If $qd^2 \leq 1$, then $k \geq 0$, so the equilibrium is stable. If $qd^2 > 1$, the trace vanishes at

$$A_H = \frac{d(1+qd^2+d)}{qd^2-1}, \qquad B_H = A_H + d.$$

With d, q fixed and $B = A + d$, $d\tau/dA = (qd^2-1)/(d(1+qd^2)) > 0$. Thus the eigenvalues cross the imaginary axis transversely at A_H, with frequency $\sqrt{c}$. Nondegeneracy of the nonlinear Hopf bifurcation requires an additional cubic-coefficient check; the trapping argument below establishes periodic motion throughout the unstable-equilibrium regime without assuming that check.

Nullclines and a trapping region. In $x > 0$,

$$\dot{x} = 0: \quad y = \frac{(B-x)(1+qx^2)}{x}, \qquad \dot{y} = 0: \quad y = \frac{A(1+qx^2)}{x}.$$

The positive quadrant is forward invariant because $\dot{x} = B > 0$ on $x = 0$ and $\dot{y} = A > 0$ on $y = 0$. For $d > 0$, choose

$$M > B, \quad 0 < \varepsilon < d, \quad Y > \max\left\{\varepsilon, \max_{\varepsilon \leq x \leq M} \frac{A(1+qx^2)}{x}\right\}.$$

Consider the polygon with vertices

$$(0,0),\ (M,0),\ (M,Y),\ (\varepsilon,Y),\ (0,Y-\varepsilon).$$

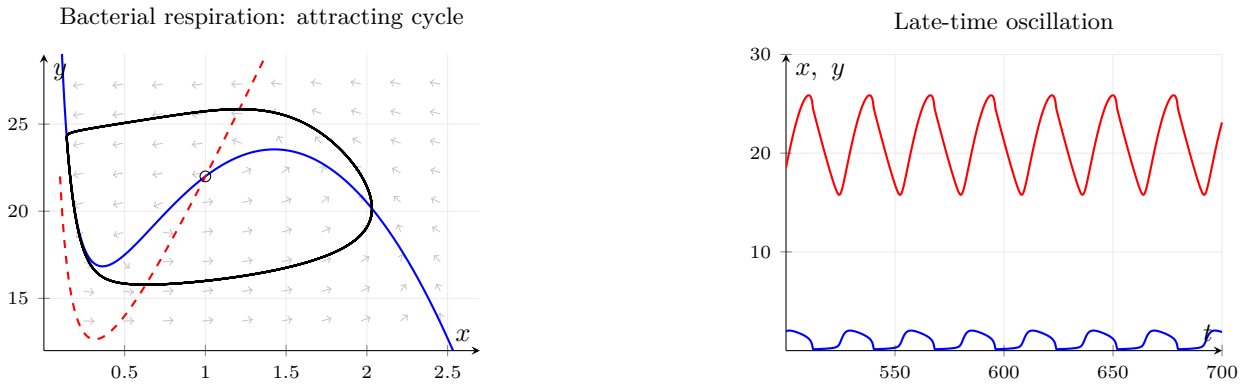
On its bottom and left edges the field points inward. On $x = M$, $\dot{x} \leq B - M < 0$. On $y = Y$ with $x \geq \varepsilon$, $\dot{y} = A - H(x)Y < 0$. On the short sloping edge $y - x = Y - \varepsilon$ with $0 \leq x \leq \varepsilon$,

$$\frac{d}{dt}(y-x) = x - d < 0,$$

again pointing inward. This gives a compact trapping region containing the equilibrium.

If $qd^2 > 1$ and $A > A_H$, the unique physical equilibrium is a repeller. Excise a sufficiently small outward-flowing neighborhood of it (an adapted ellipse, not necessarily a Euclidean circle). The remaining annulus traps trajectories and contains no equilibria. The Poincaré–Bendixson theorem then gives periodic orbits; an attracting periodic orbit occurs between the inner repeller and the outer inward boundary. These are sufficient conditions for sustained oscillations.

For example, $q = 10$, $A = 2$, $B = 3$ give $d = 1$, $c = 1/11$, $\tau = 6/11 > 0$, and $\Delta = 1/11$. Numerical integration approaches a stable limit cycle. Its approximate coordinate ranges are $0.142 < x < 2.032$ and $15.78 < y < 25.86$. The plot below shows the nullclines and the attracting cycle. For $B \le A$, no positive periodic orbit can exist: integrating $\dot{x} - \dot{y} = d - x$ over a period would require the positive mean of x to equal $d \le 0$.



Parameters: $A = 2$, $B = 3$, $q = 10$. In the phase plane blue is $\dot{x} = 0$, red dashed is $\dot{y} = 0$, and black is the attracting cycle. In the time plot blue is x and red is y.

MATLAB: `ps4_8_2_10.m`.

□

8.2.12 (Analytical criterion to decide if a Hopf bifurcation is subcritical or supercritical) Any system at a Hopf bifurcation can be put into the following form by suitable changes of variables:

$$\dot{x} = -\omega y + f(x, y), \qquad \dot{y} = \omega x + g(x, y),$$

where f and g contain only higher-order nonlinear terms that vanish at the origin. As shown by Guckenheimer and Holmes (1983, pp. 152–156), one can decide whether the bifurcation is subcritical or supercritical by calculating the sign of the following quantity:

$$\begin{aligned} 16a = {} & f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy} \\ & + \frac{1}{\omega}\left[f_{xy}(f_{xx} + f_{yy}) - g_{xy}(g_{xx} + g_{yy}) - f_{xx}g_{xx} + f_{yy}g_{yy}\right] \end{aligned}$$

where the subscripts denote partial derivatives evaluated at $(0, 0)$. The criterion is: If $a < 0$, the bifurcation is supercritical; if $a > 0$, the bifurcation is subcritical.

a) Calculate a for the system $\dot{x} = -y + xy^2$, $\dot{y} = x - x^2$.

b) Use part (a) to decide which type of Hopf bifurcation occurs for $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ at $\mu = 0$. (Compare the results of Exercises 8.2.2–8.2.4.)

(You might be wondering what a measures. Roughly speaking, a is the coefficient of the cubic term in the equation $\dot{r} = ar^3$ governing the radial dynamics at the bifurcation. Here r is a slightly transformed version of the usual polar coordinate. For details, see Guckenheimer and Holmes (1983) or Grimshaw (1990).)

Solution. **(a)** Here $\omega = 1$, $f(x, y) = xy^2$, and $g(x, y) = -x^2$. The relevant derivatives at $(0, 0)$ are

$$f_{xxx} = 0, \quad f_{xyy} = 2, \quad g_{xxy} = g_{yyy} = 0,$$

$$f_{xx} = f_{xy} = f_{yy} = 0, \qquad g_{xx} = -2, \quad g_{xy} = g_{yy} = 0.$$

Every quadratic-product term in the stated criterion is therefore zero. Hence

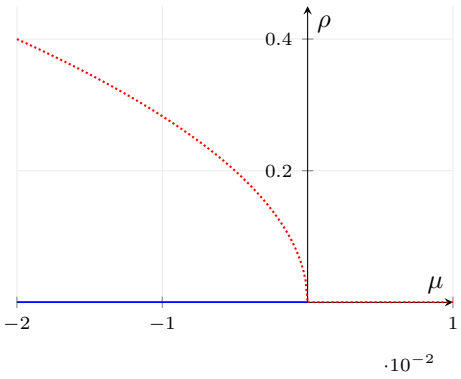
$$16a = 2, \qquad a = \frac{1}{8} > 0.$$

(b) The parameter-dependent linearization has eigenvalues $\mu \pm i$, so it crosses the imaginary axis with nonzero speed at $\mu = 0$. Since $a > 0$, the Hopf bifurcation is subcritical. Its local radial equation is

$$\dot{\rho} = \mu\rho + \frac{1}{8}\rho^3 + \cdots .$$

Thus for small $\mu < 0$ there is an unstable limit cycle surrounding the stable origin, with $\rho \sim \sqrt{-8\mu}$. It contracts into the origin as $\mu \uparrow 0$; for $\mu > 0$ the origin is unstable and there is no nearby small cycle. This confirms the numerical and heuristic conclusions in 8.2.3–8.2.4. The MATLAB file repeats the numerical portraits and the amplitude diagram.

Subcritical Hopf amplitude diagram



Solid: stable equilibrium amplitude. Dotted: unstable equilibrium or unstable-cycle amplitude. The cycle branch is the leading local normal form, not an exact global amplitude formula.

MATLAB: `ps4_8_2_12.m`.

□

8.3.1 (Brusselator) The Brusselator is a simple model of a hypothetical chemical oscillator, named after the home of the scientists who proposed it. (This is a common joke played by the chemical oscillator community; there is also the "Oregonator," "Palo Altonator," etc.) In dimensionless form, its kinetics are

$$\dot{x} = 1 - (b + 1)x + ax^2y,$$

$$\dot{y} = bx - ax^2y$$

where $a, b > 0$ are parameters and $x, y \geq 0$ are dimensionless concentrations.

a) Find all the fixed points, and use the Jacobian to classify them.

b) Sketch the nullclines, and thereby construct a trapping region for the flow.
c) Show that a Hopf bifurcation occurs at some parameter value $b = b_c$, where b_c is to be determined.
d) Does the limit cycle exist for $b > b_c$ or $b < b_c$? Explain, using the Poincaré-Bendixson theorem.
e) Find the approximate period of the limit cycle for $b \approx b_c$.

Solution. **(a)** Adding the equations yields $\dot{x} + \dot{y} = 1 - x$. Therefore the unique equilibrium is $(1, b/a)$. The Jacobian is

$$J(x,y) = \begin{pmatrix} -(b+1) + 2axy & ax^2 \\ b - 2axy & -ax^2 \end{pmatrix}, \qquad J(1, b/a) = \begin{pmatrix} b-1 & a \\ -b & -a \end{pmatrix}.$$

Thus

$$\tau = b - 1 - a, \qquad \Delta = a > 0, \qquad \lambda_{1,2} = \frac{b - 1 - a \pm \sqrt{(b-1-a)^2 - 4a}}{2}.$$

The equilibrium is stable for $b < 1 + a$ and unstable for $b > 1 + a$. It is a spiral when $|b - 1 - a| < 2\sqrt{a}$ and a node when $|b - 1 - a| > 2\sqrt{a}$, with repeated-eigenvalue nodes at equality. At $b = 1 + a$ the linearization has eigenvalues $\pm i\sqrt{a}$; the nonlinear system must be examined to classify the bifurcation.

(b) In the positive quadrant the nullclines are

$$\dot{x} = 0: \quad y = \frac{(b+1)x - 1}{ax^2}, \qquad \dot{y} = 0: \quad y = \frac{b}{ax}.$$

Set

$$L = \frac{1}{b+1}, \qquad H = \frac{b(b+1)}{a}, \qquad R = 2 - L.$$

The trapping polygon has vertices $(L, 0)$, $(R + H, 0)$, (R, H), and (L, H). Its edges have the following inward-pointing checks:

$x = L$	$\dot{x} = aL^2 y \geq 0$
$y = 0$	$\dot{y} = bx > 0$
$y = H$	$\dot{y} = bx[1 - (b+1)x] \leq 0 \quad (x \geq L)$
$x + y = R + H$	$\dot{x} + \dot{y} = 1 - x < 0 \quad (x \geq R > 1).$

The equilibrium lies inside because $L < 1 < R$ and $0 < b/a < H$. When it is a repeller, removing a small outward-flowing ellipse around it gives an equilibrium-free trapping annulus. One must use such an adapted neighborhood; an arbitrary small circle need not have outward flow everywhere.

(c) The trace is zero at

$$b_c = 1 + a, \qquad \omega_c = \sqrt{a}, \qquad \frac{d \operatorname{Re} \lambda}{db} = \frac{1}{2}.$$

For completeness, the cubic coefficient is negative. At $b = b_c$, use $u = x - 1$ and $w = -\sqrt{a}\,[x + y - 1 - b_c/a]$. Then

$$\dot{u} = -\sqrt{a}\, w + (1-a)u^2 - 2\sqrt{a}\, uw - au^3 - \sqrt{a}\, u^2 w, \qquad \dot{w} = \sqrt{a}\, u.$$

Applying the criterion in 8.2.12 gives $16 a_{\mathrm{H}} = -2a - 4$, so

$$a_{\mathrm{H}} = -\frac{a+2}{8} < 0.$$

Therefore the Hopf bifurcation is supercritical.

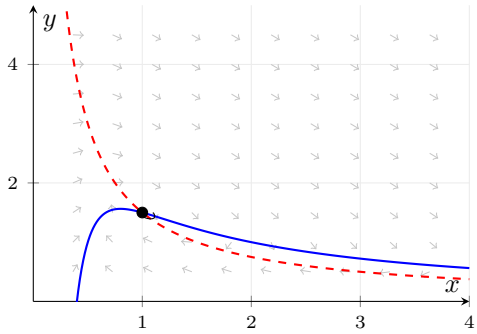
(d) For $b > b_c$, the equilibrium is unstable and the trapping annulus implies a periodic orbit by Poincaré–Bendixson. Near b_c, the supercritical Hopf calculation gives a stable small limit cycle on precisely this side. The annulus argument alone establishes existence, not uniqueness of all cycles for arbitrary parameters.

(e) As $b \downarrow b_c$ from above, the angular frequency tends to $\sqrt{a}$, so the period satisfies

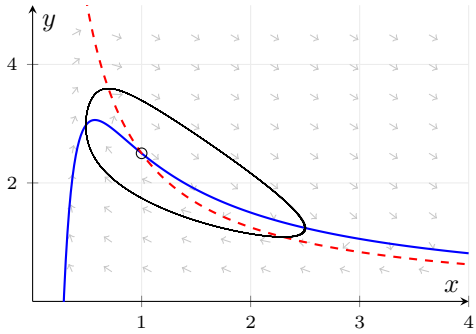
$$T \sim \frac{2\pi}{\sqrt{a}}.$$

For the illustrated choice $a = 1$, $b_c = 2$ and the near-onset period is approximately 2π.

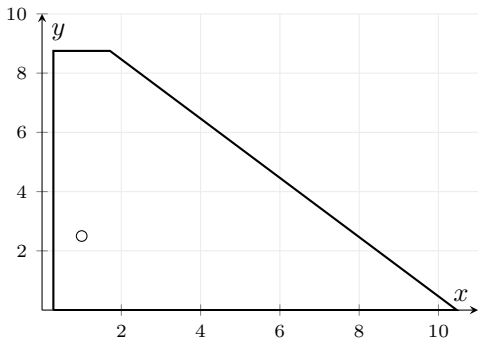
$a = 1,\ b = 1.5$



$a = 1,\ b = 2.5$



Trapping polygon: $a = 1,\ b = 2.5$



Blue curves are $\dot{x} = 0$, red dashed curves are $\dot{y} = 0$. Black trajectories approach the stable equilibrium below onset and the stable cycle above onset. The polygon is shown separately so its scale does not obscure the cycle.

MATLAB: `ps4_8_3_1.m`.

□

8.4.4 (Second-order phase-locked loop) Using a computer, explore the phase portrait of

$$\ddot{\theta} + (1 - \mu \cos\theta)\dot{\theta} + \sin\theta = 0$$

for $\mu \geq 0$. For some values of μ, you should find that the system has a stable limit cycle. Classify the bifurcations that create and destroy the cycle as μ increases from 0.

Solution. Let $v = \dot{\theta}$. On the phase cylinder $\theta \pmod{2\pi}$,

$$\dot{\theta} = v, \qquad \dot{v} = -(1 - \mu\cos\theta)v - \sin\theta.$$

The equilibria are $(0, 0)$ and $(\pi, 0)$ modulo 2π. The Jacobian is

$$J(\theta, v) = \begin{pmatrix} 0 & 1 \\ -\mu v \sin\theta - \cos\theta & -1 + \mu\cos\theta \end{pmatrix}.$$

At $(0,0)$,

$$\tau = \mu - 1, \quad \Delta = 1, \qquad \lambda_{1,2} = \frac{\mu - 1 \pm \sqrt{(\mu-1)^2 - 4}}{2}.$$

Thus it is stable for $0 \le \mu < 1$ and unstable for $\mu > 1$. At $(\pi, 0)$, $\Delta = -1$, so it is a saddle for every $\mu \ge 0$.

Creation of the cycle. At $\mu = 1$, the origin has eigenvalues $\pm i$. Put $x = \theta$, $y = -v$; the cubic terms at this parameter are

$$\dot{x} = -y, \qquad \dot{y} = x - \frac{1}{6}x^3 - \frac{1}{2}x^2 y + O(\|(x,y)\|^5).$$

The criterion of 8.2.12 gives $16 a_{\mathrm{H}} = -1$, hence $a_{\mathrm{H}} = -1/16 < 0$. The transverse crossing speed is $1/2$, so a supercritical Hopf bifurcation at $\mu = 1$ creates a stable oscillatory (librating) cycle for $\mu > 1$. Close to onset its amplitude satisfies $\rho \sim \sqrt{8(\mu - 1)}$.

Destruction of the librating cycle. Increasing μ enlarges the cycle until it meets the saddle's separatrices. To locate the connection numerically, integrate the unstable branch from $(-\pi, 0)$ forward to $\theta = 0$, and the stable branch from $(\pi, 0)$ backward to $\theta = 0$. If their positive velocities there are $v_u(\mu)$ and $v_s(\mu)$, solve

$$v_u(\mu) - v_s(\mu) = 0.$$

Starting a distance 10^{-6} from each saddle in its eigendirection and refining the shooting parameter gives

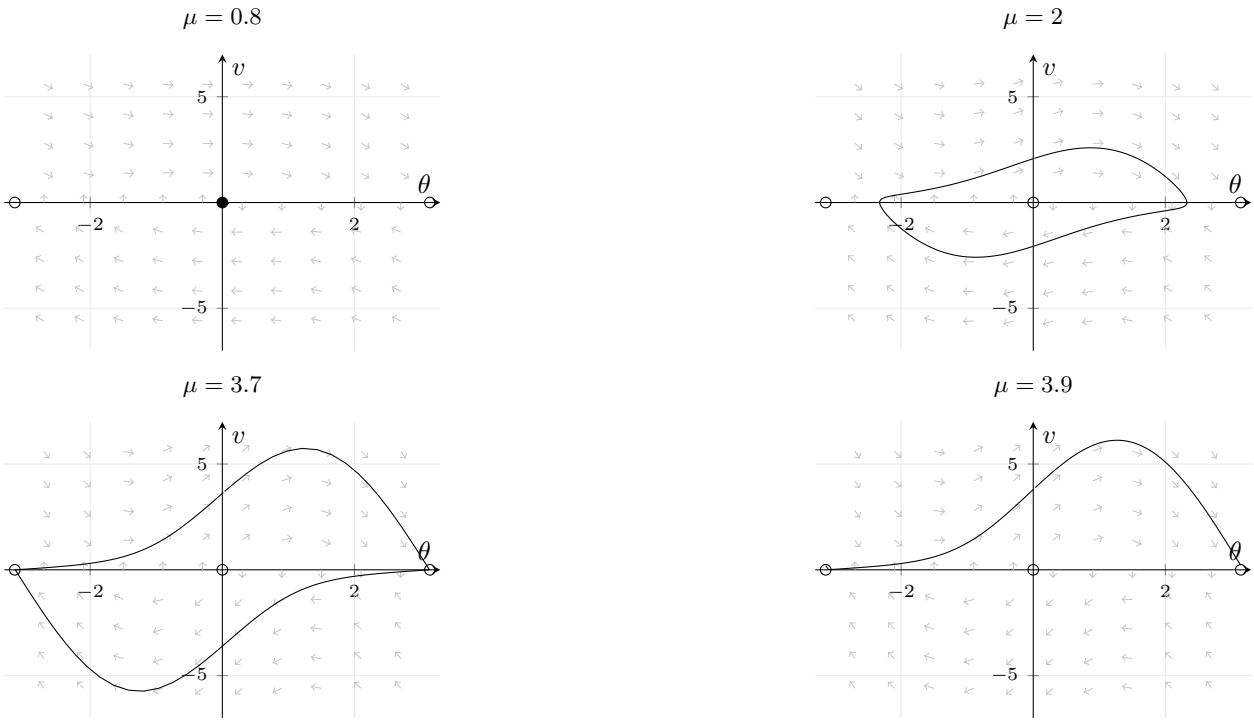
$$\mu_{\mathrm{hom}} \approx 3.72487.$$

A smaller starting offset provides a check on this value; the displayed precision is numerical, not exact. The libration period grows without bound on approach to the connection. In the unwrapped (θ, v) plane the upper connection runs from $(-\pi, 0)$ to $(\pi, 0)$, with a symmetric lower connection. On the cylinder these endpoints represent the same saddle, so this is a homoclinic bifurcation. It destroys the librating cycle. Numerical trajectories above threshold can instead approach rotating solutions; the disappearance here concerns the libration, not all possible periodic motions on the cylinder.

The energy identity

$$E = \frac{1}{2}v^2 + 1 - \cos\theta, \qquad \dot{E} = -(1 - \mu\cos\theta)v^2$$

explains the mechanism: for $\mu \le 1$ there is no energy injection, while $\mu > 1$ introduces negative damping near $\theta = 0$. The phase portraits below use $\mu = 0.8, 2, 3.7, 3.9$; trajectories are wrapped into $[-\pi, \pi]$ for the cylindrical portrait.



The left and right edges represent the same angular position. Above the homoclinic threshold, wrapping the angular coordinate splits a rotating trajectory at the seam; these discontinuities are plotting cuts, not jumps in the physical flow.

MATLAB: `ps4_8_4_4.m`.

□

8.5.4 (Logistic equation with sinusoidal harvesting) In Exercise 3.7.3 you were asked to consider a simple model of a fishery with constant harvesting. Now consider a generalization in which the harvesting varies periodically in time, perhaps due to daily or seasonal variations. To keep things simple, assume the periodic harvesting is purely sinusoidal (Benardete et al. 2008). Then, if the fish population grows logistically in the absence of harvesting, the model is given in dimensionless form by

$$\dot{x} = rx(1-x) - h(1+\alpha\sin t).$$

Assume that $r, h > 0$ and $0 < \alpha < 1$.

a) Show that if $h > r/4$ the system has no periodic solutions, even though the fish are being harvested periodically with period $T = 2\pi$. What happens to the fish population in this case?

b) By using a Poincaré map argument like that in the text, show that if $h < \dfrac{r}{4(1+\alpha)}$, there exists a 2π-periodic solution—in fact, a stable limit cycle—in the strip $1/2 < x < 1$. Similarly, show there exists an unstable limit cycle in the strip $0 < x < 1/2$. Interpret your results biologically.

c) What happens in between cases (a) and (b), i.e., for $\dfrac{r}{4(1+\alpha)} < h < \dfrac{r}{4}$?

Solution. Write $F(t,x) = rx(1-x) - h(1+\alpha\sin t)$ and $T = 2\pi$. The nonautonomous equation becomes an autonomous flow on the cylinder by adding $\dot{\theta} = 1$, $\theta = t \pmod{2\pi}$.

(a) A periodic solution would satisfy, after averaging one period,

$$0 = r\langle x(1-x)\rangle - h = \frac{r}{4} - h - r\left\langle \left(x - \frac{1}{2}\right)^2 \right\rangle.$$

Thus $h \le r/4$ is necessary. For $h > r/4$, no periodic solution exists. Moreover, for any initial $x(0) \ge 0$, integration gives

$$x(t) \le x(0) + \left(\frac{r}{4} - h\right) t + h\alpha(\cos t - 1).$$

The right side becomes negative, so the population must hit zero in finite time. Biologically this means extinction; the differential equation's negative continuation is not a physical population model. At $h = r/4$, periodicity would force $x \equiv 1/2$, which cannot solve the equation when $\alpha > 0$, so there is no periodic solution at equality either.

(b) Suppose $h < r/[4(1+\alpha)]$. On the boundaries of the upper strip,

$$F(t, 1/2) \ge r/4 - h(1+\alpha) > 0, \qquad F(t,1) = -h(1+\alpha\sin t) < 0.$$

Hence $[1/2, 1]$ is forward invariant. Its time-T map P is continuous, strictly increasing, and satisfies $P(1/2) > 1/2$, $P(1) < 1$. It therefore has a fixed point, giving a periodic solution entirely in $1/2 < x < 1$. Its multiplier is

$$P' = \exp\left(\int_0^{2\pi} r[1 - 2x(t)]\, dt\right) < 1,$$

so this cycle is stable. On the lower strip $0 \le x \le 1/2$, the field points outward in forward time because $F(t,0) < 0$ and $F(t,1/2) > 0$. The backward-time map preserves that strip and has a fixed point. This gives a periodic solution in $0 < x < 1/2$ with $P' > 1$, hence an unstable cycle.

The upper periodic population is sustainable and attracting. The lower periodic population is a time-dependent survival threshold: populations below it at the same seasonal phase eventually go extinct, whereas those above it approach the upper cycle.

(c) In the intermediate range the outcome depends on the parameters; neither endpoint estimate alone decides it. The two periodic solutions persist until a saddle-node of cycles at a critical harvesting level

$$\frac{r}{4(1+\alpha)} < h_c(r,\alpha) < \frac{r}{4}.$$

For $h < h_c$ there are two positive periodic solutions (one stable and one unstable); at $h = h_c$ there is a semistable periodic solution with multiplier 1; for $h > h_c$ there is no positive periodic solution and every nonnegative population eventually hits zero. To justify the fold structure, on every interval where the time-T flow exists, its first and second variations satisfy

$$U' = F_x U, \quad U(0) = 1, \qquad V' = F_x V - 2rU^2, \quad V(0) = 0.$$

Thus $P' > 0$ and $P'' < 0$, so the graph of $P(x) - x$ is strictly concave and has at most two zeros. Increasing h strictly decreases P by the scalar comparison principle. The two zeros therefore merge tangentially and disappear. The strict lower bound follows because the two cycles are still hyperbolic at the lower endpoint (the isolated boundary tangencies do not give constant boundary solutions); the strict upper bound follows from the averaged identity above and persistence of hyperbolic cycles.

For a numerical way to find h_c, put $x = 1/2 + u'/(ru)$. The Riccati equation becomes

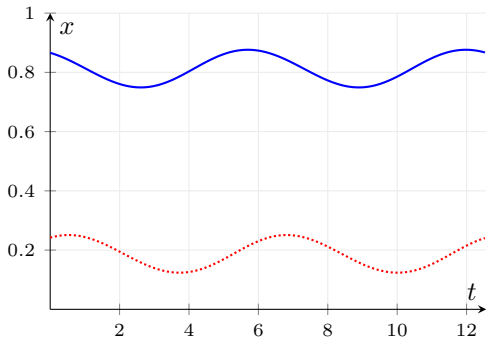
$$u'' = \left[\frac{r^2}{4} - rh(1+\alpha\sin t)\right] u.$$

Integrate its two fundamental solutions for one period to obtain the monodromy matrix $M(h)$. Its determinant is 1. The fold is the first value of h at which $\operatorname{tr} M(h) = 2$ and there is a positive periodic u; this is the principal Floquet threshold. For $r = 1$ and $\alpha = 1/2$, bisection gives

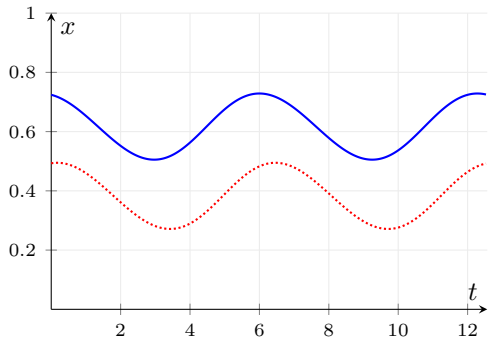
$$h_c \approx 0.2426848, \qquad \frac{r}{4(1+\alpha)} = \frac{1}{6}, \qquad \frac{r}{4} = \frac{1}{4}.$$

Thus, for example, $h = 0.23$ is in the intermediate range but still has the two periodic solutions, whereas $h = 0.247$ is beyond the fold. The MATLAB file computes the threshold and plots stable/unstable seasonal population cycles.

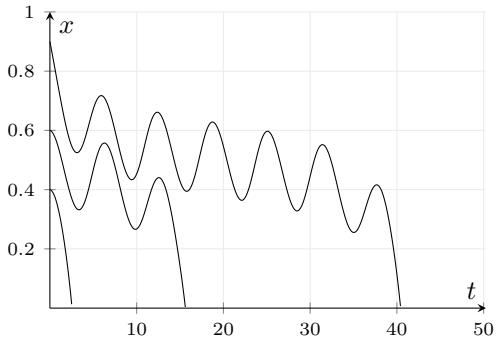
$r = 1$, $\alpha = 1/2$, $h = 0.15$



$r = 1$, $\alpha = 1/2$, $h = 0.23$



$r = 1$, $\alpha = 1/2$, $h = 0.247$



Blue solid curves are stable seasonal populations; red dotted curves are unstable seasonal thresholds. The middle plot lies in the intermediate parameter range. In the last plot no periodic population remains, and the forward trajectories terminate at extinction.

MATLAB: `ps4_8_5_4.m`.

□